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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/326,180	12/14/2011	Ben A. Barres	STAN-457CON	7835

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EXAMINER
GUDIBANDE, SATYANARAYAN R

ART UNIT	PAPER NUMBER
1676	

NOTIFICATION DATE	DELIVERY MODE
02/06/2015	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 13/326,180	Applicant(s) BARRES ET AL.	
	Examiner SATYANARAYANA R. GUDIBANDE	Art Unit 1676	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☒ Responsive to communication(s) filed on *12/17/2014*.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL**. 2b)☒ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5)☒ Claim(s) *1-15* is/are pending in the application.

5a) Of the above claim(s) *12-14* is/are withdrawn from consideration.

6)☐ Claim(s) _____ is/are allowed.

7)☒ Claim(s) *1-11 and 15* is/are rejected.

8)☐ Claim(s) _____ is/are objected to.

9)☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10)☒ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date. _____ .

4) ☐ Other: _____.

The present application is being examined under the pre-AIA first to invent provisions.

DETAILED ACTION

Election/Restrictions

Applicant's election without traverse of group I, election of Alzheimer's as species of disease condition and election of anti-C1q antibody as the complement inhibitor in the reply filed on 12/17/2014 is acknowledged.

Information Disclosure Statement

The listing/citing of references in the specification is not a proper information disclosure statement. 37 CFR 1.98(b) requires a list of all patents, publications, or other information submitted for consideration by the Office, and MPEP § 609.04(a) states, "the list may not be incorporated into the specification but must be submitted in a separate paper." Therefore, unless the references have been cited by the examiner on form PTO-892, they have not been considered. Several patent and non-patent literature citations have been used in the instant application throughout the entire specification. An IDS has not been filed in the application along with the legible copies of the citations for the office to consider.

Specification

The disclosure is objected to because of the following informalities: the specification has special characters/symbols that are not art recognizable, for example (CT□) in [113] and elsewhere. Amendment to specification filed on 03/09/2012 has addressed few of these

special characters but there are several that need to be addressed. Appropriate correction is required.

Status of pending claims

Applicant's amendment to claims in the response filed on 12/17/2014 has been acknowledged.

Claims 1-15 are pending.

Claims 12-14 withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected invention, there being no allowable generic or linking claim. Election was made **without** traverse in the reply filed on 12/17/2014.

Claims 1-11 and 15 are examined on the merit.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claim 8 rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

In the instant invention, applicants claim a method of inhibiting synapse loss comprising the step of: administering a therapeutic amount of a complement inhibitor to a patient suffering

from adverse synapse loss (claim 1). Claim 8 recite a limitation ‘wherein said inhibitor upregulates expression of a complement inhibitor in neural cells’. It is unclear how an inhibitor that was administered to inhibit a 'complement inhibitor' would ‘upregulates the expression of a complement inhibitor’. The two limitations are contradictory to one another and hence does not further limit the subject matter as claimed in the base claim.

The following is a quotation of 35 U.S.C. 112(d):

> (d) REFERENCE IN DEPENDENT FORMS.—Subject to subsection (e), a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), fourth paragraph:

> Subject to the [fifth paragraph of 35 U.S.C. 112 (pre-AIA)], a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

Claim 8 is rejected under 35 U.S.C. 112(d) or pre-AIA 35 U.S.C. 112, 4th paragraph, as being of improper dependent form for failing to further limit the subject matter of the claim upon which it depends, or for failing to include all the limitations of the claim upon which it depends. In the instant invention, applicants claim a method of inhibiting synapse loss comprising the step of: administering a therapeutic amount of a complement inhibitor to a patient suffering from adverse synapse loss (claim 1). Claim 8 recite a limitation ‘wherein said inhibitor upregulates expression of a complement inhibitor in neural cells’. It is unclear how an inhibitor that was administered to inhibit a 'complement inhibitor' would ‘upregulates the expression of a complement inhibitor’. The two limitations are contradictory to one another and hence does not

further limit the subject matter as claimed in the base claim. Applicant may cancel the claim(s), amend the claim(s) to place the claim(s) in proper dependent form, rewrite the claim(s) in independent form, or present a sufficient showing that the dependent claim(s) complies with the statutory requirements.

The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

> (a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

> The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

Claims 1-11 and 15 are rejected under 35 U.S.C. 112, first paragraph, as failing to comply with the enablement requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to enable one skilled in the art to which it pertains, or with which it is most nearly connected, to make and/or use the invention.

In the instant invention applicants a method of inhibiting synapse loss comprising the steps of administering a therapeutic amount of a complement inhibitor to a patient suffering from adverse synapse loss.

The specification as disclosed does not enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to practice the invention without undue burden of experimentation as the specification lacks specific working examples in treating neurodegenerative disease such as Alzheimer's disease (AD).

The factors to be considered in determining whether a disclosure meets the enablement requirements of 35 U.S.C. 112, first paragraph, have been described in *In re Wands*, 858 F.2d 731, 8 USPQ2d 1400 (Fed. Cir., 1988). The court in Wands states, "Enablement is not precluded by the necessity for some experimentation, such as routine screening. However, experimentation needed to practice the invention must not be undue experimentation. The key word is 'undue', not 'experimentation'" (Wands, 8 USPQ2sd 1404). Clearly, enablement of a claimed invention cannot be predicated on the basis of quantity of experimentation required to make or use the invention. "Whether undue experimentation is needed is not a single, simple factual determination, but rather is a conclusion reached by weighing many factual considerations" (Wands, 8 USPQ2d 1404). Among these factors are: (1) the nature of the invention; (2) the breadth of the claims; (3) the state of the prior art; (4) the predictability or unpredictability of the art; (5) the amount of direction or guidance presented; (6) the presence or absence of working examples; and (7) the quantity of experimentation necessary.

While all of these factors are considered, a sufficient amount for a *prima facie* case is discussed below.

(1) The nature of the invention and (2) the breadth of the claims:

The claims are drawn to a method of inhibiting synapse loss comprising the steps of administering a therapeutic amount of a complement inhibitor to a patient suffering from adverse synapse loss.

The invention as claimed does not identify the chemical nature of the complement inhibitor, i.e., whether the inhibitor is a small organic molecule or a peptide or a nucleic acid, etc., and the claims as recited is drawn to administering a therapeutic amount of an unknown

compound to a patient suffering from adverse synapse loss. The patient as it is recited in the dependent claims is suffering from central or peripheral nervous disorder and the disorder is identified as 'Alzheimer's disease' (claim 5) and the inhibitor has been identified as C1q inhibitor (claim 11). The invention as claimed encompasses any and all known and unknown complement inhibitors whose administration would inhibit synapse loss in a patient who is suffering from adverse synapse loss due to any and all neurodegenerative disorders.

(3) The state of the prior art and (4) the predictability or unpredictability of the art:

The state of the prior art is discussed below with regards to treating a patient who is suffering from adverse synapse loss due to Alzheimer's disorder by administering a therapeutic amount of a complement inhibitor.

Choi, 2003, J Korean Neurol Assoc., 21, 513-520, states that endogenous complement inhibitors in the brain may protect against the neuroinflammation in Alzheimer's disease based on the study in which $A\beta_{1-42}$ and TNF-α upregulated the expression of C1 inhibitor in neuroblastoma cells, in a cell culture study (Abstract).

The instant specification applicants disclose that the cause for Alzheimer's disease (AD) is unknown and environmental factors are the focus of active investigation [51].

Veerhuis, 1998, Acta Neuropathol, 96, 287-296 discloses that activation of locally produced complement factors may act as a mediators between amyloid deposits and neurodegenerative changes in AD as suggested by in situ and in vitro studies. Veerhuis also discloses that C1 inhibitor found to be secreted in a functionally active form by astrocytes cultured from postmortem human brain specimens as well as neuroblastoma cell lines (Abstract).

Yasojima, 1999, Brain Research, 833, 297-301 states that complement regulators C1 inhibitor and CD59 do not significantly inhibit complement activation in AD (Title).

Sahu, Immunopharmacology 49 (2000) 133–148, discloses that "[T]he exact risk associated with complement inhibition **is not known**, primarily because of a paucity of sufficient data. It is known, however, that complement deficiencies are associated with increased susceptibility to infection, increased susceptibility to endotoxin as a result of impaired clearance, and with autoimmunity such as that seen in systemic lupus erythematosus and glomerulonephritis. Thus, in principle, these effects may be perceived as risks" (page 142, column 1, paragraph 1).

Johnson, Experimental Neurology 138, 198–205 (1996) discloses an in vitro study of Rabbit anti-rat C1q antibodies wherein injured rat brain sections were exposed the aforementioned antibody (page 199, columns1-2). Johnson observed weak C1q immuno staining in the hippocampal formation, including presubicular areas contiguous to the entorhinal cortex (page 200, column 2, last paragraph). The data only supported that the complement system generally participates in response to brain injury as well as in AD (abstract).

The afore-illustrated scenarios are clearly indicative of the fact that AD is a neurodegenerative disorder that is not well understood as the cause for the disease condition itself is unknown. The studies of Choi and Veerhuis has been the results of in vitro studies that indicate that endogenously produced complement factors may be responsible for AD and C1 inhibitor expression has been observed to be secreted in active form in cultured cells derived from postmortem brain specimens. Yasojima has observed that C1 inhibitor and CD59 do not significantly inhibit complement activation in AD. Sahu clearly states that "exact risk associated

with complement inhibition **is not known**". Hence the afore-illustrated studies imply that there is certain amount of unpredictability in the art with regards to understanding the cause of AD and the role of C1 inhibitor and the complement system believed to be the cause of AD symptoms. Johnson concluded that complement system generally participates in response to brain injury as well as in AD.

(5) The amount of direction or guidance presented and (6) the presence or absence of working examples:

The instant invention is drawn to treating inhibiting synapse loss in a patient who is suffering from adverse synapse loss, i.e., AD by administering a therapeutic amount of a complement inhibitor. The instant specification does not have any specific example of an in vitro or an in vivo study related to treating AD. The specification is silent with regards to guidance necessary for one of ordinary skill in the art to practice the invention for treating synapse loss in AD patients.

(7) The quantity of experimentation necessary:

Considering the state of the art and the unpredictability associated with the art as afore-discussed combined with the lack of any specific examples to practice the invention, one of ordinary skill in the art would be burdened with quantity of experimentation to practice the instant invention. Over and above the chemical nature of the complement inhibitor is not clearly evident in the invention as recited and one of ordinary skill in the art would be burdened with testing innumerable compounds to treat a disease condition such as AD the cause for which is not clearly identified according to the instant specification.

It is the Office's position that one skilled in the art could not practice the instant invention without undue experimentation. It is also noted, considering the *a priori* unpredictability in the art as discussed above with regard to AD as a disease condition and treatment of the same, testing the efficacy of innumerable number of compounds of the instant invention to treat AD as a disease condition is not enabled.

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the claims at issue are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on a nonstatutory double patenting ground provided the reference application or patent either is shown to be commonly owned with this application, or claims an invention made as a result of activities undertaken within the scope

of a joint research agreement. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO internet Web site contains terminal disclaimer forms which may be used. Please visit http://www.uspto.gov/forms/. The filing date of the application will determine what form should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to http://www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

1. Claims 1-4 and 7-10 are rejected on the ground of nonstatutory double patenting as being unpatentable over claim 1 of U.S. Patent No. 8148330. Although the claims at issue are not identical, they are not patentably distinct from each other because in the instant invention applicants a method of inhibiting synapse loss comprising the steps of administering a therapeutic amount of a complement inhibitor to a patient suffering from adverse synapse loss. The claim of U.S. Patent No. 8148330 is drawn to treating glaucoma a disorder from optic nerve damage or loss to the field of vision is a species of the generic disease condition recited in the instant application as a patient suffering from adverse synapse loss. Hence species of disorder 'glaucoma' of U.S. Patent No. 8148330 anticipates instant invention.

2. Claims 1-11 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-9 of copending Application No. 13586556. Although the claims at issue are not identical, they are not patentably distinct from each other

because in the instant invention applicants a method of inhibiting synapse loss comprising the steps of administering a therapeutic amount of a complement inhibitor to a patient suffering from adverse synapse loss. The claims of copending Application No. 13586556 are drawn to a method of treating aging-associated cognitive decline in a patient, comprising: administering a therapeutic amount of a complement inhibitor to a patient suffering from cognitive decline. The disease condition 'aging-associated cognitive decline' is a species of the generic disease condition recited in the instant application as a patient suffering from adverse synapse loss. Hence species of disorder 'aging-associated cognitive decline' of copending Application No. 13586556 anticipates instant invention.

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Satyanarayana R. Gudibande whose telephone number is 571-272-8146. The examiner can normally be reached on M-F 8-4.30.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Karlheinz Skowronek can be reached on 571-272-9047. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free).

/SATYANARAYANA R GUDIBANDE/
Primary Examiner, Art Unit 1676